

Koustubh Kulkarni

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SUMMARY

Computer Science undergraduate specializing in **AI/ML systems and Python backend development**. Experience in building **data pipelines, automation workflows, and intelligent systems** using LLMs and machine learning. Interned at **NVIDIA** and **KPMG**, working on scalable architectures and enterprise AI solutions. Strong problem-solving background as an **ACM ICPC Regionalist** and **CodeChef 3*** programmer. **Interests & goals:** Intelligent automation, backend systems, and scalable data-driven architectures for real-world applications.

EDUCATION

- KLE Technological University** — B.E. Computer Science (CGPA: **8.7/10**) 2022 – 2026

EXPERIENCE

- KPMG** Bangalore
 - Jan 2026 – Present* Digital Lighthouse Intern
 - Engineered **LLM-driven AI agents** using **LangChain/RAG** integrated with **Snowflake** for enterprise data retrieval and workflow automation.
 - Developed modular **data processing pipelines** and orchestration workflows using LangGraph, improving efficiency and cost optimization of inference.
 - Trained in enterprise-scale **Java, Spring Boot, and Cloud-native** application design.
- NVIDIA** Hubli/Remote
 - Aug 2024 – Jun 2025* Project Intern
 - Built a deep learning pipeline for **HDR reconstruction** using a dual-branch **U-Net** for multi-exposure image fusion. Trained over 60K+ images dataset including SDR and IR images
 - Developed **virtual camera systems** using **VirtIO** for real-time streaming between Linux and Android VMs. Resulting in optimized **QEMU and FFmpeg pipelines**, reducing streaming latency by **30%**.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL
Core: Pytoech,Pandas, NumPy, REST APIs, FastAPI, Data Pipelines, JSON/CSV Processing, Vector DBs, CI/CD
AI & Systems: ML, LangChain, LangGraph, RAG, PostgreSQL, MongoDB, Snowflake, Docker, MCP

PROJECTS

- CreekPay — Blockchain BNPL Platform (Hack Karnataka 2nd Place)** 2025
 - Python ,FastAPI, OCR, ML* GitHub
 - Led **AI/ML and backend development** for a P2P BNPL platform, building a credit scoring engine using Gradient Boosting with financial feature engineering.
 - Designed real-time credit evaluation API and database schemas for secure transaction handling.
 - Collaborated in a 3-member team to integrate ML services with blockchain-based transaction logging and frontend systems, ensuring **secure, scalable, and modular system design**.
- FinMitraAI — AI Financial Advisor** 2025
 - Python, FastAPI, LangChain, Groq APIs* GitHub
 - Built an **LLM-powered financial advisor** delivering personalized investment insights with **85% accuracy**.
 - Implemented real-time **market analytics pipelines** processing 1,000+ financial data points.
 - Deployed a scalable FastAPI backend integrated with **Groq APIs** for low-latency inference.
- Synk-App — Real-Time Chat Application** 2025
 - Node.js, Socket.IO, MongoDB* GitHub
 - Developed a secure WebSocket-based messaging platform with JWT authentication and persistent storage.
 - Implemented concurrent session handling and scalable backend event management.

ACHIEVEMENTS

- ACM ICPC Regionalist (2025) | CodeChef 3*** (500+ DSA problems solved)
- WiDS Datathon Top 100 Global | Hack Karnataka 2nd Place** among 100+ teams